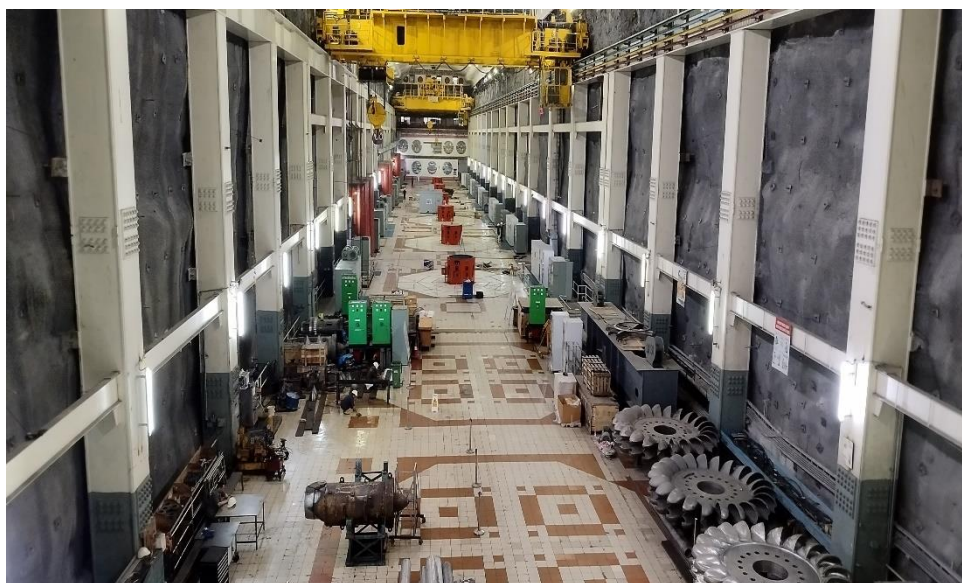




Annual Maintenance Completion Report (2025-2026)

Generating unit V



Mechanical Maintenance Unit

Tala Hydropower Plant

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1. Overview

The Annual Maintenance of Unit 5 was carried out simultaneously with Unit 6, covering essential inspection and servicing activities to ensure reliable operation of the generating unit. The major work planned was the replacement of the MIV service seals; however, this could not be undertaken due to time constraints and has been deferred to the next Annual Maintenance. Replacement of the runner was another major activity planned for the unit.

Other works carried out included pre-testing and overhauling of nozzles, replenishment of nitrogen in the MIV accumulator banks, calibration of MIV pressure gauges, cleaning of the Main Heat Exchanger, and inspection of the Cooling Water pumps. In addition, Ultrasonic Testing was conducted on the MIV decompression pipeline, bypass pipeline, and the pressure accumulators of both the MIV and governor systems.

All activities were executed in coordination with the concerned sections, ensuring adherence to safety and quality requirements.

Scheduled vs Actual date of start and completion of maintenance on the generating unit is shown in Table-1

SN	Unit #	Approved Schedule			Actual Completion		
		Start	End	No of Days	Start	End	No of Days
1.	Unit V	24.01.2026	11.03.2026	38	05.01.2026	24.02.2026	39

Table 1: Unit 5 AM schedule

The Annual Maintenance work of the unit commenced on 05.01.2026. The unit was put to mechanical spin at 05:15 hours on 25.02.2026, dry run and then shut down upon reaching the temperature limit of the bearings. The synchronization of the unit was further delayed due to nozzle no.2 position failure. The machine was successfully synchronized after the replacement of the nozzle at 19:38 hours on 02.03.2026.

The major maintenance activities conducted for the unit are detailed hereunder and all test/calibration reports, checklists and inspection notes are placed as Annexure.

2. Turbine

Table 2: Spares consumed for Turbine system

S. No	Materials	Quantity
1	H.T Allen bolt, M20x150 H/thread, Gr.12.9	5
2	SS Hex headed bolt & nut M10x60mm, half	24
3	H.T Allen bolt, M16x160, grade 12.9,	2
4	NBR Endless O-ring 320x6mm 70 shore	14
5	NBR Endless O-ring 246x6mm 70 shore	2

6	NBR Endless O-ring 335x6mm 70 shore	4
7	NBR Endless O-ring 19.5x3mm 70 shore	4
8	NBR Endless O-ring 28x3mm 70 shore	2
9	Cup seal P.P-200x230x16mm	2
10	Cup Seal P.P-110X130X12mm	4
11	Wiper seal PU with casing110X120X7/10mm,	2
12	PTFE ring seal200X209.2X7.5mmwith inner,	2
13	Wiper seal PU with casing200X220X12/16mm	3
14	Locking nut KMTA 18 for nozzle	1
15	SS Allen bolt, M36x340, grade 9.8,	8

2.1. Nozzle Injector Assembly

All five nozzle injector assemblies were subjected to functional checks through in-situ oil pressure testing. During the test, conducted at an initial pressure of 100 bar for 30 minutes, a pressure drop was observed in Nozzle 2, which was subsequently dismantled for overhaul as part of the Annual Maintenance. Additionally, Nozzle 4 was dismantled, having exceeded the recommended four-year overhaul interval, with over five years of operation. The remaining nozzle assemblies that did not require dismantling underwent inspection of the needle tips and liner assemblies, and fasteners were replaced wherever necessary.

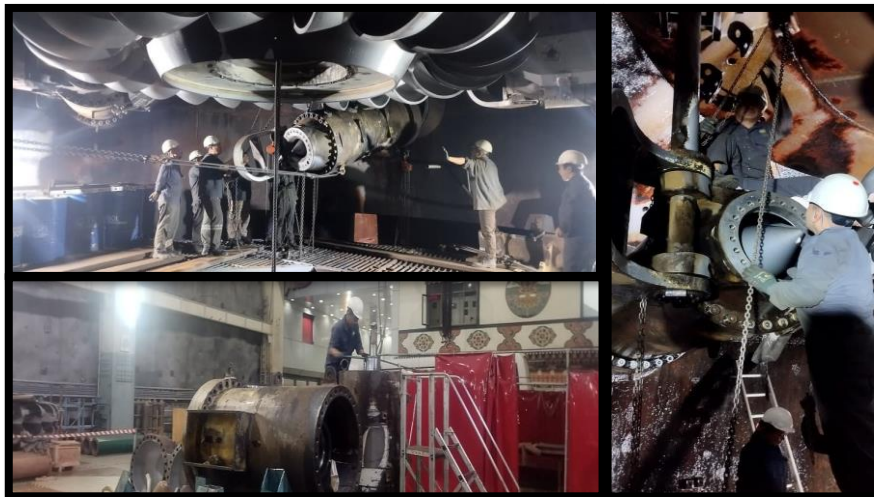


Figure 1: Overhauling & servicing of nozzle injectors

2.2 Runner

Table 3: Runner particulars

Particulars	Runner (3020)
Make	Andritz Hydro, Switzerland
Type	Forged Fabricated and hard coated
Hard Coating	HVOF
Cumulative Running Hours	71,322.19 hours as of 05.01.2026

Repair/ reclamation History

Reclaimed once at BHSL; between October 2017 and November 2019 after 37,149.34 running hours

The unit VI runner 3020 was in operation from 2012 and its running hours since the commission has reached 71,322.19 hours by the time the unit was taken under shutdown for annual maintenance on 05.01.2026.



Figure 2: Erosion observed on runner buckets

The physical condition of the runner buckets was visually inspected, and erosion was observed on all buckets. Considering the significant extent of erosion and that the runner had already operated for more than 71,000 hours, it was decommissioned and replaced with reclaimed runner 2020.

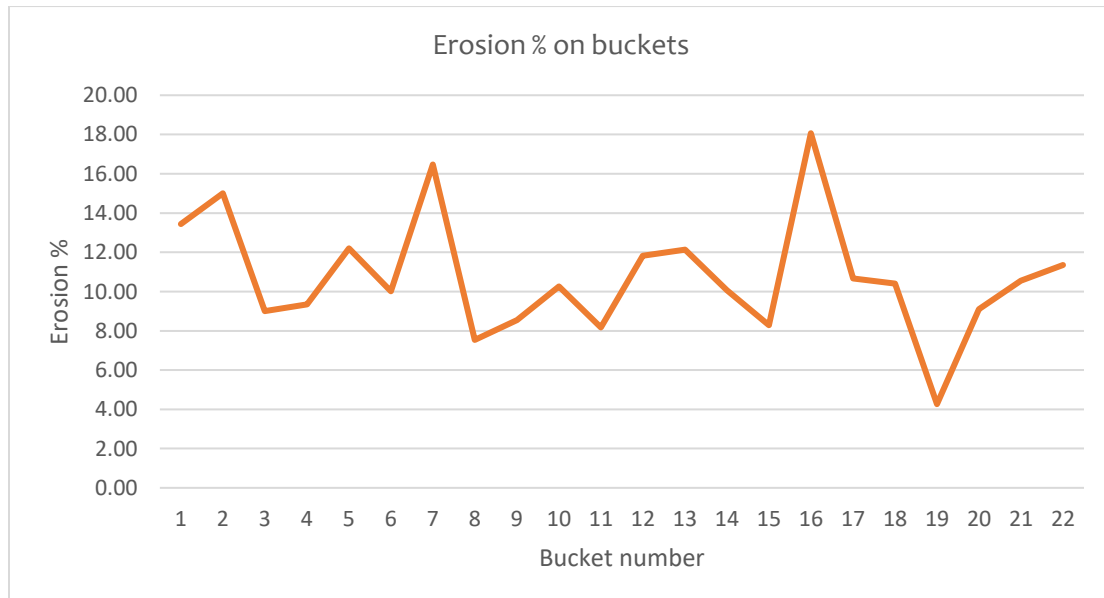


Figure 3: Erosion rates

The above chart gives information on parent material erosion on each bucket of the runner. The erosion is recorded maximum of 18.07% on bucket 16 while the least is observed on bucket 19 whose erosion is about 4.26%.

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The runner will be dispatched for reclamation at BHSL.



Figure 4: Runner 3020 removed from unit 5

3. Governor and MIV System

Table 4: Spares consumed for Governor and MIV system

S. No	Description	Quantity
1	MIV control pressure duplex filter element	1
2	SS pipe Tee 100mm NB, Sch 120	1
3	SS 90 Deg elbow 100NB pipe, Sch 120	1
4	4" NB Seamless Seamless SS pipe	4
5	S.S flange 100NB for S.S pipe,	2
6	Rubber clamps, 80 mm	4

3.1 Governing System

The main function of governor system is to maintain the rotational speed of turbine within acceptable limits during no-load condition. It performs this function by sensing changes in speed and accordingly controlling flow of water into turbine at fast rate to maintain the speed with reference to grid frequency.

For this unit, the SCADA implementation had been successfully completed during the previous Annual Maintenance. Accordingly, no works related to the HMC or NCC were required this year, unlike in past maintenance cycles where such activities were a major part of the scope. Instead, routine servicing tasks were carried out to ensure system reliability. These included thorough cleaning of the OPU tank, inspection of pump and oil filters, and replenishment of nitrogen pressure in the accumulator bank to maintain optimal system performance. All high-pressure pipelines associated with VTA (Valve Table Assembly) were carefully inspected for any signs of leakage, and necessary tightening or corrective measures were taken wherever required.

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3.1.1 Valve Table Assembly (VTA)



Figure 5: Valve Table Assembly

The VTA is a critical component of the hydraulic control system, primarily responsible for directing and regulating oil flow to the servomotors of the needle and deflector systems. It houses the solenoid valves, pressure gauges, and other control elements that interface with the SCADA system for remote operation and monitoring.

With the SCADA integration completed during the previous Annual Maintenance, the VTA now facilitates more precise and efficient control of the turbine's hydro-mechanical functions, reducing the need for manual intervention and improving overall system responsiveness.

Deflector gap setting was done and the set result is tabulated below.

Table 5: Deflector gap setting readings

Nozzle	Initial reading (mm)	Final reading (mm)
J1	7.5	7.5
J2	5	8
J3	8.5	8
J4	16	8
J5	13	8

3.2 MIV System

The MIV system is the primary isolation for turbine and principal protection for the station against penstock water.

The pressure gauges of MIV control panel and accumulator have been calibrated at HRDC and four gauges had pressure deviations beyond acceptable limits. These gauges will be replaced after the receipt of new gauges. The report is attached.

In order to ensure safe parameters of the control pipelines for safe and smooth operations, periodic NDT is highly recommended to be done annually since these are considered among the most critical parts.

The ultrasonic testing of MIV bypass pipeline, decompression pipeline and pressure accumulators for thickness verification was done by an inspection team from HRDC to inspect any possible deviations in the pipeline thickness or any crack development before it amplifies. The test result has been attached.

The servo valve of the decompression pipeline was opened and inspected for debris. This activity is recommended during every annual maintenance and also during runner inspections to ensure

that no foreign particles or sediments obstruct the valve operation, which could affect the proper functioning of the decompression system and lead to abnormal pressure conditions.



Figure 6: Inspection on MIV decompression line

OPU pump capacity was checked and found normal. Functionality check for internal leakages and servicing was carried on all the directional control solenoid valves provided in OPU and MIV control panel. Contactors and limit switches of MIV were checked for their healthiness and replacement done wherever necessary. During the functionality check, no N₂ gas leakage was observed.

4. Cooling Water System

4.1. Clean Cooling Water (CCW) System

During this Annual Maintenance, CCW pump B underwent a complete inspection with dismantling, cleaning, and replacement of critical components such as the mechanical seals, bearings and glands.

Following the overhauling, servicing and reassembly, shaft alignment between the pump and motor was carried out using an Easy-Laser alignment system. The laser alignment process ensured accurate correction of both angular and parallel misalignments, thereby minimizing vibration and mechanical load on the coupling. Final alignment readings were recorded and confirmed to be within acceptable tolerance limits. The alignment of pump A was also checked.



Figure 7: Overhauling and servicing of CCW pump

4.2. Raw Cooling Water (RCW) System

The pumps were visually inspected. During the inspection, the bearing hubs and pump casings were cleaned to remove any accumulated debris or contaminants. For pump 2, the oil tank was cleaned and the oil hose was replaced. The pump NRV was replaced. The motorized strainer and cyclone separator were also thoroughly cleaned to ensure smooth filtration and uninterrupted flow.



Figure 8: Cleaning of strainer filter elements

These maintenance activities were carried out as preventive measures to maintain system performance and reliability.



Figure 9: Overhauling of RCW pump

After completion of the annual maintenance of Unit 5, overheating was observed in RCW Pump 1 during unit spinning. The pump also exhibited abnormal vibration and noise during operation. In coordination with EMU, the pump was overhauled for inspection, and components such as the end-cap bearing and thrust bearing were replaced.

However, the issue persisted even after the replacement. An official from HRDC was subsequently called to carry out vibration testing, which confirmed that the source of the vibration was the motor. The rectification process is still underway; meanwhile, the unit has been synchronized and brought into operation, with the standby pump being operated for the time being.

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4.3. Heat Exchanger

Each unit is equipped with a shell-and-tube type heat exchanger (MHE) to facilitate the circulation of clean cooling water for the stator air coolers, OPU coolers, and bearing coolers. Raw cooling water is supplied through the tubes inside the heat exchanger to enable heat transfer.

During the annual maintenance, the heat exchanger was opened, and the tubes were thoroughly cleaned and flushed to remove any scaling or deposits that could impede heat transfer efficiency. After cleaning, the end covers were reassembled, and pressure testing was conducted to ensure there were no leaks and that the unit was fit for continued operation. Leakage was detected in 20 tubes at the inlet side and 16 tubes at the outlet side. These tubes were plugged to ensure continued safe operation.



Figure 10: Cleaning of MHE

5. Internal/ external services availed

- Runner inspection (Magnetic Particle Inspection) by HRDC
- Ultrasonic testing on decompression, bypass pipelines, pressure accumulators and distributor pipeline joints by HRDC
- Vibration analysis of CCW and RCW pumps by HRDC
- Calibration of pressure gauges on MIV panels by HRDC
- Testing of Governor and MIV system oil samples by HRDC

6. Maintenance Challenges

- Overheating and abnormal vibration in RCW Pump 1 during unit start-up, requiring coordination with EMU and HRDC for inspection and ongoing rectification; standby pump operated meanwhile.

7. Any Modification and improvement

- The jockey pump installed earlier in the MIV OPU system was commissioned during this Annual Maintenance by CPU team. The jockey pump is provided to maintain and retain the required oil pressure in the OPU system.

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Annexure

S.NO	Annexure	Description
1	Annexure 1	Auxiliary Final Checklist
2	Annexure 2	Deflector mechanical gap setting
3	Annexure 3	Deflector servo timing set
4	Annexure 4	Fasteners tightness checklist
5	Annexure 5	G100 Governor system functionality check
6	Annexure 6	Governor system final checks
7	Annexure 7	Governor nitrogen pressure retention check
8	Annexure 8	Governor OPU pump capacity check
9	Annexure 9	Governor OPU safety valve setting
10	Annexure 10	Governor system pre-checklist
11	Annexure 11	Governor system pre-checklist
12	Annexure 12	Governor system pressure retention check
13	Annexure 13	Inspection of Decompression valve
14	Annexure 14	MIV system final checks
15	Annexure 15	MIV OPU safety valve setting
16	Annexure 16	MIV pre-checklist
17	Annexure 17	MIV system pressure retention checklist
18	Annexure 18	MIV system timing checklist
19	Annexure 19	Nozzle assembly checklist
20	Annexure 20	Nozzle post assembly checklist
21	Annexure 21	Nozzle pre-checklist
22	Annexure 22	Nozzle timing checklist
23	Annexure 23	Turbine Final checklist
24	Annexure 24	Underwater components Inspection
25	Annexure 25	UT report on MIV high-pressure pipelines
26	Annexure 26	UT report on pressure accumulators
27	Annexure 27	Pressure gauge calibration report
28	Annexure 28	Pump Vibration assessment report

Submitted by:



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